Social Network Analysis

LINK ANALYSIS

PathSim: Measuring Similarity of Objects in Heterogeneous Network

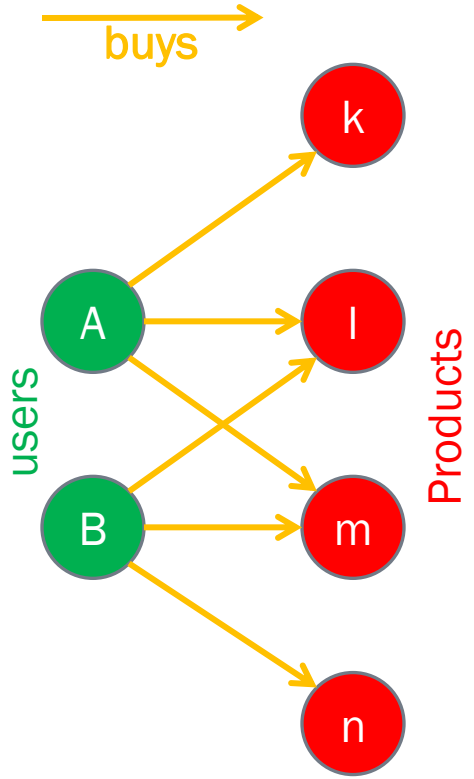
Heterogeneous Networks

- A tuple of the form $(V, E, \mathcal{A}, \mathcal{R}, \varphi, \psi)$ represents an information networking system if
  - V is the set of vertices
  - E is the set of edges
  - $\mathcal{A}$ is the set of different node types present in the network
  - $\mathcal{R}$ is the set of different link types present in the network
  - $\varphi(v): V \rightarrow \mathcal{A}$ maps each vertex to a node type
  - $\psi(e): E \rightarrow \mathcal{R}$ maps each edge to a link type
- If $|\mathcal{A}| = 1$ as well as $|\mathcal{R}| = 1$, then the system is termed as a homogeneous network
- On the contrary, if $|\mathcal{A}| > 1$ or $|\mathcal{R}| > 1$, or both, then the system is termed as a heterogeneous network

Heterogeneous Networks: Variants

- When $|\mathcal{A}| > 1$ and $|\mathcal{R}| = 1$, then we have a heterogeneous network consisting of vertices of more than one types, and only one types of links

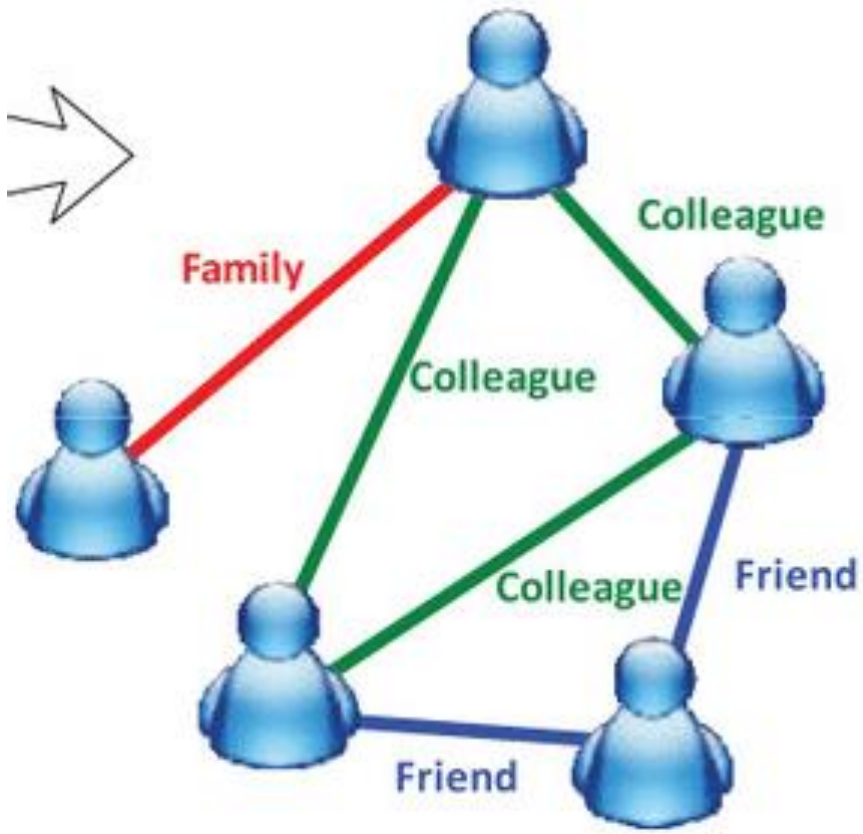
- A typical example is consumer-product purchase network, where
  - $\mathcal{A} = \{users, products\}$, and
  - $\mathcal{R} = \{user \longrightarrow products | user\ buys\ product\}$



Heterogeneous Networks: Variants

- When $|\mathcal{A}| = 1$ and $|\mathcal{R}| > 1$, then we have a heterogeneous network consisting of vertices of one type, but there are more than one type of links between these vertices

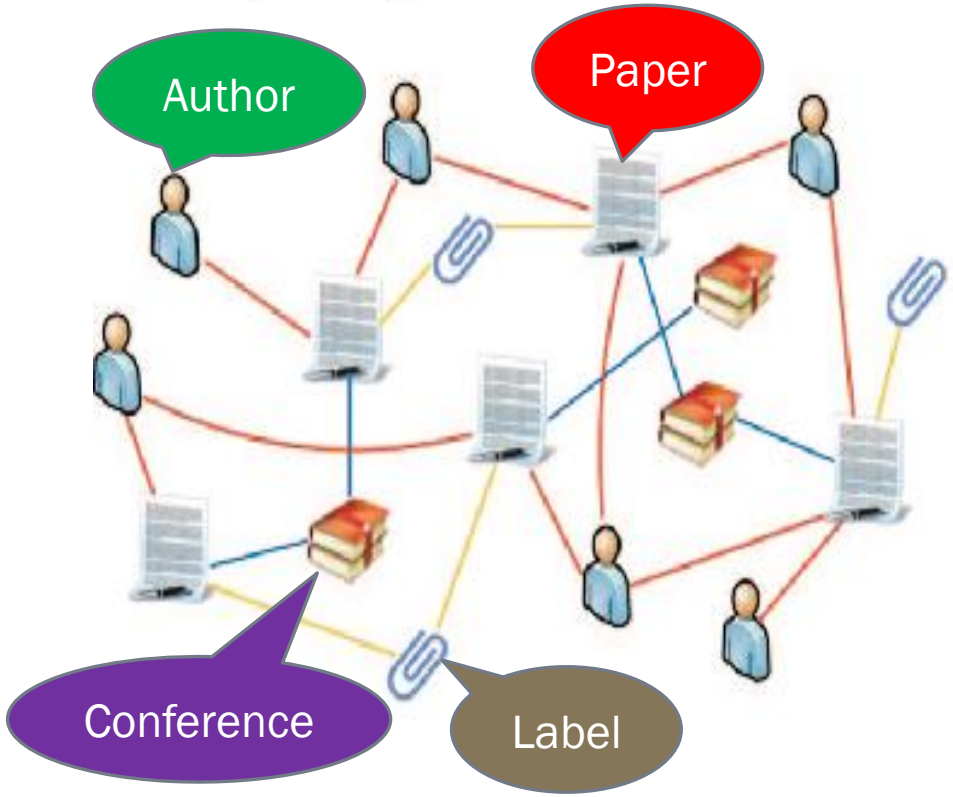
- A typical online social networking platform;
  - only one type of vertices, viz. users of the network;
  - There are more than one type of links: friends in real life, family members in real life, office colleague in real life, and so on.



Liu and Yu [2019]

Heterogeneous Networks: Variants

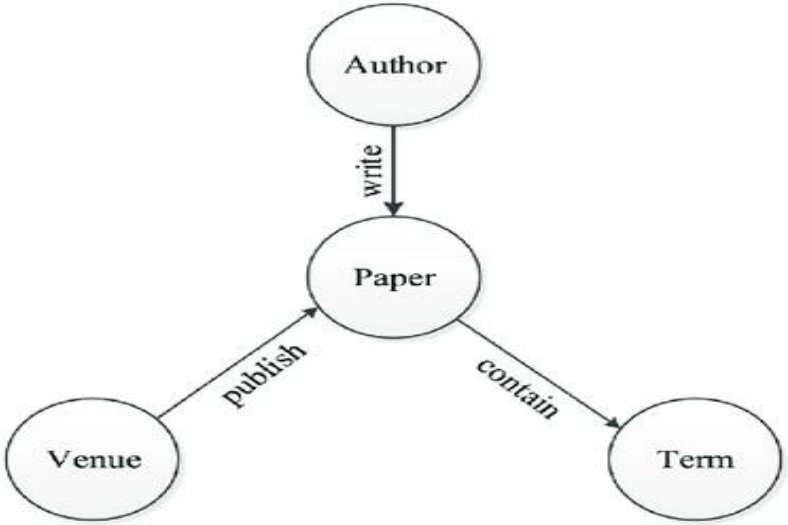
- When both $|\mathcal{A}| > 1$ and $|\mathcal{R}| > 1$, then we have a heterogeneous network consisting of vertices and links of more than one type
- A typical bibliographic network consisting of authors, papers, conference venues, etc., and various kinds of relationship between these entities



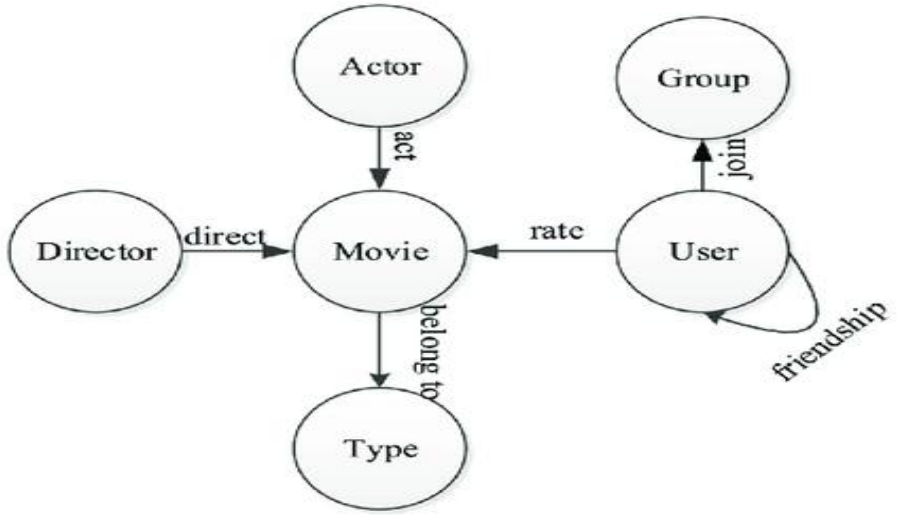
https://www.semanticscholar.org/paper/HRank%3A-A-Path-based-Ranking-Framework-in-Network-Li-Shi/186d8239daa10cedb7be946387a9326a0a3c9999

Heterogeneous Networks: Network Schema

- A meta-data level outline for a heterogeneous directed network $G(V, E)$ and the information tuple $(V, E, \mathcal{A}, \mathcal{R}, \varphi, \psi)$, where $\varphi: V \rightarrow \mathcal{A}$ is the object type mapping, and $\psi: E \rightarrow \mathcal{R}$ is the link type mapping. The corresponding network schema is given by $T_G = (\mathcal{A}, \mathcal{R})$



(A) DBLP network with a star network schema



(B) Douban Movie network with a general network schema

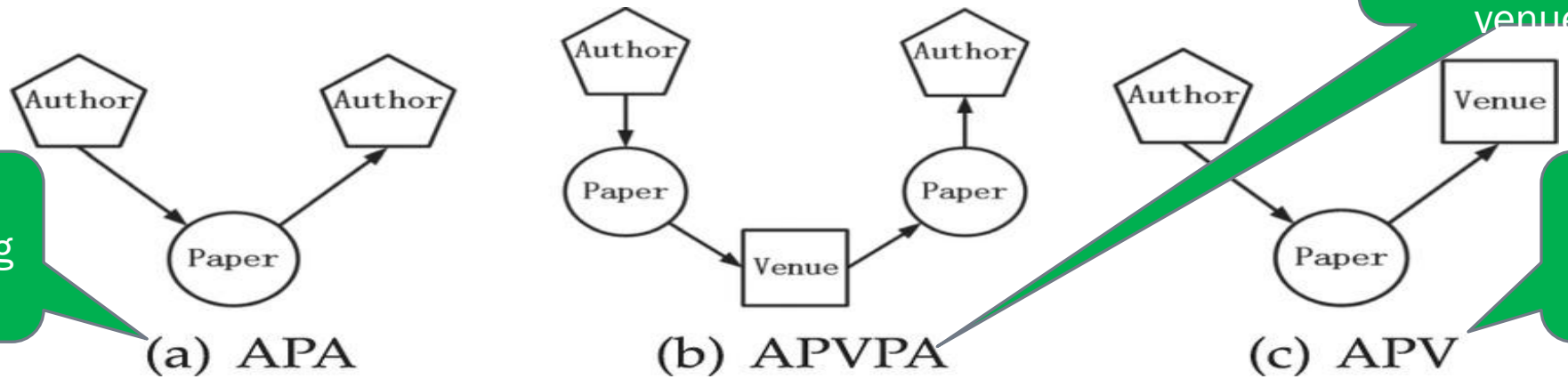
https://www.researchgate.net/publication/314129795_Generic_network_schema_agnostic_sparse_tensor_factorization_for_single-pass_clustering_of_heterogeneous_information_networks

Heterogeneous Networks: Meta-Path

- A meta-path is a meta-level description of the structural connectivity between the entities
- Different paths deliver varying semantic similarity/differences or measure different topological connectivity
- A meta-path is a path $\mathcal{P}$ of length ℓ defining a composite relation over the ℓ links $\mathcal{R} = \mathcal{R}_1 \circ \mathcal{R}_2 \circ \mathcal{R}_3 \circ \cdots\cdots\cdots \circ \mathcal{R}_\ell$ and $\ell + 1$ objects $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3, \cdots\cdots\cdots, \mathcal{A}_{\ell+1}$ denoted in the form

$$\mathcal{A}_1 \xrightarrow{\mathcal{R}_1} \mathcal{A}_2 \xrightarrow{\mathcal{R}_2} \mathcal{A}_3 \cdots\cdots\cdots \mathcal{A}_\ell \xrightarrow{\mathcal{R}_\ell} \mathcal{A}_{\ell+1}$$

authors publishing papers in same venue



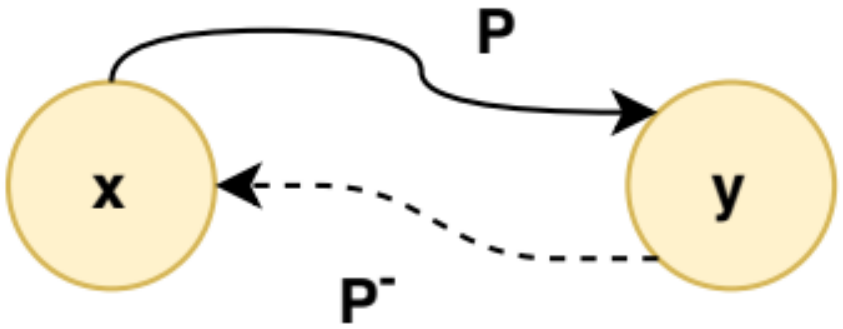
authors collaborating in a paper

author and venue connected via some paper

https://www.researchgate.net/figure/Example-for-Meta-path-in-HIN-on-the-bibliographic-network-2-Figure-3-defines-the-meta_fig1_339302745

PathSim: Formulation

- A measure of similarity search scoring and ranking in heterogeneous information networks
- Use the notion of meta-paths for the formulation
- A meta-path of the form $\mathcal{P} = (\mathcal{P}_\ell, \mathcal{P}_\ell^-)$ where the starting and ending object is the same, is termed as a round-trip meta-path. By default, it is always symmetric.



PathSim: Formulation

- A meta-path based symmetric similarity measure, PathSim, between two objects x and y of the same type can be given as follows:

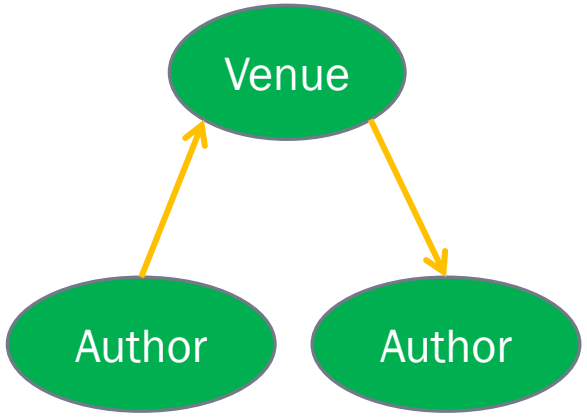
$$s(x,y) = \frac{2 \times |\{p_{x \rightsquigarrow y} | p_{x \rightsquigarrow y} \in \mathcal{P}\}|}{|\{p_{x \rightsquigarrow x} | p_{x \rightsquigarrow x} \in \mathcal{P}\}| + |\{p_{y \rightsquigarrow y} | p_{y \rightsquigarrow y} \in \mathcal{P}\}|}$$

here $p_{x \rightsquigarrow y}$ is path instance between x and y, and $p_{x \rightsquigarrow x}$ and $p_{y \rightsquigarrow y}$ are roundtrip path instances

- The salient features of PathSim
  - Symmetric: $s(x,y) = s(y,x)$
  - Normalized: $s(x,y) \in [0,1]$
  - Self-Maximized: $s(x,x) = 1$

PathSim: Illustration

The table below depicts the venue based publication frequency of some authors. To find the author most similar to Mike



Author	MOD	VLDB	ICDE	KDD
Mike	2	1	0	0
Jim	50	20	0	0
Mary	2	0	1	0
Bob	2	1	0	0
Ann	0	0	1	1

PathSim: Illustration

The visibility V_p of individual authors:

$$V_p(Mike) = 2 \times 2 + 1 \times 1 + 0 \times 0 + 0 \times 0 = 5$$

$$V_p(Jim) = 50 \times 50 + 20 \times 20 + 0 \times 0 + 0 \times 0 = 2900$$

$$V_p(Mary) = 2 \times 2 + 0 \times 0 + 1 \times 1 + 0 \times 0 = 5$$

$$V_p(Bob) = 2 \times 2 + 1 \times 1 + 0 \times 0 + 0 \times 0 = 5$$

$$V_p(Ann) = 0 \times 0 + 0 \times 0 + 1 \times 1 + 1 \times 1 = 2$$

The overall connectivity C_p between Mike and other authors are as follows:

$$C_p(Mike, Jim) = 2 \times 50 + 1 \times 20 + 0 \times 0 + 0 \times 0 = 120$$

$$C_p(Mike, Mary) = 2 \times 2 + 1 \times 0 + 0 \times 1 + 0 \times 0 = 4$$

$$C_p(Mike, Bob) = 2 \times 2 + 1 \times 1 + 0 \times 0 + 0 \times 0 = 5$$

$$C_p(Mike, Ann) = 2 \times 0 + 1 \times 0 + 0 \times 1 + 0 \times 1 = 0$$

PathSim: Illustration

Similarity scores in terms of V_p and C_p are as follows

$$s(Mike, Jim) = \frac{2 \times 120}{5 + 2900} = 0.0826$$

$$s(Mike, Mary) = \frac{2 \times 4}{5 + 5} = 0.8$$

$$\boldsymbol{s(Mike, Bob) = \frac{2 \times 5}{5 + 5} = 1.0}$$

$$s(Mike, Ann) = \frac{2 \times 0}{5 + 5} = 0.0$$

PathSim: Exercise

Consider a restaurant review network containing objects of two types : restaurant (R) and user (U). There exists a review (V) relationship between U and R as shown in below table, where each cell shows the number of reviews given by a user to a restaurant. The task is to find the peer restaurant for Mint.

Author	Michelle	Alice	Bob	Eve
Mint	2	4	0	0
Pavilion	4	0	2	1
Symposium	2	4	0	0
Sky Route	0	0	1	3